TNA in Nutrition: A Throughput-Based Model of Metabolic Coherence and Collapse

Claudio Bresciano Independent Researcher Theorem of Axiomatic Necessity (TNA)

Abstract

Nutrition is traditionally modeled as energy balance, macronutrient ratios, or biochemical pathways. This paper proposes an alternative framework based on the Theorem of Axiomatic Necessity (TNA), viewing metabolism as an open non-equilibrium system where coherence requires continuous throughput (Ψ) to discharge incoming incoherence (G). When G exceeds Ψ, accumulated incoherence (ℵ) manifests as obesity, inflammation, or deficiency disorders. Empirical phenomena—from obesity epidemics to longevity under caloric restriction—are reframed as throughput saturation rather than primary deficits. The model is falsable, predictive, and compatible with existing nutritional data, offering a unified explanation for metabolic persistence and failure.

Keywords: Theorem of Axiomatic Necessity (TNA); metabolic throughput; incoherence accumulation; obesity; caloric restriction; non-equilibrium metabolism

1. Introduction: The Body as a Household Plumbing System

Consider a common household kitchen sink supplied by a water line. Nutrients enter (G), accumulate in storage (ℵ), and are processed/discharged through metabolism (Ψ). If input exceeds processing, storage overflows (fat); if discharge is blocked, the system backs up (inflammation).

The human body operates under the same structural constraint:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

where G is caloric/nutrient intake, Ψ is metabolic processing/discharge (digestion, oxidation, excretion), and ℵ is accumulated noise (adipose tissue, toxins, oxidative stress).

This paper applies TNA to nutrition, showing that many "disorders" are consequences of throughput imbalance rather than isolated pathologies.

2. Core TNA Mapping to Metabolic Systems

TNA VARIABLE	METABOLIC EQUIVALENT	BIOLOGICAL MANIFESTATION	EXAMPLE CONDITION
G (generation)	Caloric/nutrient intake	Food consumption, glucose spikes	Hyperphagia, processed foods
Ψ (throughput)	Digestion, oxidation, excretion, autophagy	Basal metabolism, exercise, glymphatic-like clearance	Reduced in sedentary lifestyle
ℵ (incoherence)	Adipose tissue, inflammation, toxins	Visceral fat, CRP levels, oxidative damage	Obesity, metabolic syndrome
F_min (minimal coherence)	Homeostatic baseline	Mitochondrial efficiency, insulin sensitivity	Violated in aging/disease

3. Specific Applications

3.1 Obesity Epidemics

G high (ultra-processed foods, sugar) > Ψ low (sedentary life). ℵ accumulates as fat. Application: Intermittent fasting increases Ψ (autophagy) —reduces ℵ without extreme G reduction (studies show 5-10% weight loss sustained).

3.2 Ketogenic Diets and Metabolic Flexibility

Reduce G (carbohydrates) → force fuel switch (fat to ketones). Initial ℵ high (keto flu). Long-term Ψ high (mitochondrial efficiency). Application: Improves insulin sensitivity 30-50% in resistant individuals (clinical data).

3.3 Caloric Restriction and Longevity

G 20-30% low → Ψ maintained high (autophagy active). ℵ low (reduced oxidative damage). Application: Correlated with +5-10 healthy years (Okinawa studies, rodent models).

3.4 Micronutrient Deficiencies

G includes cofactors for Ψ enzymatic. Lack → specific routes collapse (e.g., B12 → methylation failure). Application: Supplementation restores Ψ —reduces fatigue/depression 20-40%.

3.5 Western vs. Ancestral Diets

Western: G high incoherent (processed) → Ψ saturated (inflammation). Ancestral: G balanced → Ψ high (anti-inflammatory). Application: Mediterranean shift reduces markers 40% in months (PREDIMED trial).

3.6 Aging and Metabolic Decline

Age → Ψ low (mitochondrial slowdown). G constant → ℵ high (sarcopenia, diabetes). Application: Resistance exercise + protein boosts Ψ —delays collapse 10-20 years.

3.7 Microbiome as Extended Ψ

The gut microbiome acts as an external throughput module —ferments indigestible fiber, produces short-chain fatty acids, modulates inflammation. G high processed/low fiber → microbiome dysbiosis → Ψ reduced (leaky gut, reduced SCFA). ℵ high (chronic inflammation). Application: High-fiber/fermented foods increase microbiome Ψ —reduces systemic ℵ (studies show 20-30% lower inflammation markers). Probiotics/prebiotics as "Ψ boosters".

4. Predictive Power and Falsability

- Predicts: Interventions increasing Ψ (exercise, fasting) outperform G reduction alone for fat loss/inflammation.
- Falsable: If Ψ enhancement fails to reduce ℵ in controlled trials, model weakens.
- Distinguishes from traditional models: Not "calories in/out" —throughput mismatch.

5. Figure Suggestion: Metabolic Sacrifice Rate Curve

[For generation with Flux/SimpAI: "Curva de sacrifice rate metabólica: eje X sacrifice rate (low to high), eje Y coherence (optimal band in middle), zonas left 'accumulation collapse' (obesity/diabetes), right 'over-catabolism collapse' (cachexia). Estilo científico limpio."]

Interpretation: Optimal coherence at intermediate sacrifice (autophagy/clearance) —too low = ℵ accumulation, too high = loss of reserves.

6. Implications for Dietary Practice

- Diet as Ψ engineering: Not "restrict calories" —balance G with Ψ capacity.
- Fasting/exercise as discharge valves.
- Personalized: Measure Ψ (RMR, activity trackers) to match G.
- Prevention: Lifestyle to maintain $\Psi > G$ (sleep, movement, whole foods).

7. Conclusion

Metabolic disorders are not mysteries of biochemistry. They are throughput failures of the body.

TNA provides a unified, mechanical framework for understanding nutritional stability—from daily energy balance to lifelong health—complementary to existing biochemical and caloric models.

Canonical Statement

> *The body does not fail because it eats too much. It fails when it cannot process what it eats.*